

YASH GOYAL

• Sydney • +61 402 412 386 • yashgoyal1120@gmail.com • linkedin.com/in/yashgoyal11

PROFESSIONAL SUMMARY

Production-focused Data Scientist and ML Engineer with end-to-end experience designing, deploying and monitoring AI systems from LLM-powered agents and knowledge-graph QA pipelines to GPU-accelerated ML pipelines in live production environments. Proficient in Python, SQL, and cloud-native stacks (AWS, GCP, Azure). Strong software engineering discipline: clean code, CI/CD, GitHub-based workflows. Experienced in collaborating with cross-functional stakeholders to translate business problems into scalable, reliable solutions. Master of Computer Science (Data Science and AI) graduate from the University of Sydney, where I also serve as a Teaching Tutor in the School of Computer Science.

CORE TECHNICAL SKILLS

AI / ML / GenAI	LLMs, LangChain, LangGraph, Hugging Face Transformers, Agentic AI, RAG, PyTorch, TensorFlow, Keras, XGBoost, LightGBM, Scikit-Learn
Cloud & MLOps	Azure (AI Studio, ML), AWS, Google Cloud, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Terraform, MLflow, SAS Viya
Data & Pipelines	Python, SQL (PostgreSQL, MySQL), Pandas, NumPy, Spark (familiar), Neo4j, REST APIs, FastAPI, Flask
Engineering	Clean code practices, test-driven development, code review, Git/GitHub workflows, Agile/Scrum
Other	Computer Vision (OpenCV, YOLOv5), R, JavaScript, C++, Bash/Shell, Blockchain

PROFESSIONAL EXPERIENCE

Cloud and Technology Intern

Jan 2026 – Present

SAS Institute • Sydney, Australia

- Built and deployed real-time analytics workflows on SAS Viya (cloud-native ML platform), supporting end-to-end model operationalisation across cloud environments.
- Developed internal tooling and automation scripts that improved deployment reliability and reduced onboarding time for analytics pipelines.
- Collaborated cross-functionally with engineering and analytics teams to diagnose issues and ensure production-grade system reliability.
- Winner of SAS Viya for Learners Challenge (ANZ) 2025: implemented a full end-to-end ML analysis in Viya, placing top in the regional competition.

Machine Learning Engineer

Sep 2025 – Dec 2025

AirLabOne • Sydney, Australia

- Designed and deployed GPU-accelerated ML pipelines integrated with NVIDIA Omniverse, managing the full lifecycle from experimentation through to production.
- Built systems for real-time extraction and distribution of model insights to user endpoints, with focus on reliability, performance and scalability.
- Collaborated with hardware and software engineers to tune models and infrastructure for production-grade performance.
- Applied CI/CD and GitHub-based workflows to maintain code quality and streamline deployment iterations.

Teaching Tutor, Object-Oriented Programming (INFO1113)

Aug 2025 – Present

School of Computer Science, University of Sydney • Sydney, Australia

- Communicate complex technical concepts clearly to undergraduate students; deliver targeted written and verbal feedback.
- Designed practice problems and code-review frameworks improving student understanding of software design patterns.

Software Engineering Intern

Oct 2023 – Feb 2024

ITS Lab, IIT Madras • Chennai, India

- Built a Python/Flask backend for real-time traffic video analytics, reducing processing latency by ~25% through pipeline optimisation.
- Shipped full-stack solution with live metrics dashboard; improved user engagement by ~30%.
- Implemented secure authentication and optimised DB schema for ~15% faster query performance.

KEY PROJECTS

Agentic MLOps Monitor

Python • Azure ML • LangChain • MLflow • FastAPI • Docker • GitHub Actions

- Designed and deployed an agentic AI system on Azure ML that continuously monitors live model performance, automatically detects data drift and concept drift across production pipelines.
- Integrated an LLM layer (via LangChain) to generate natural-language diagnostic reports and remediation recommendations, surfaced through a FastAPI endpoint.
- Automated retraining triggers via GitHub Actions CI/CD pipelines when drift thresholds are breached; tracked all experiments and model versions with MLflow.

Graph-QA Agent (LLM-powered Knowledge-Graph Question Answering)

Neo4j • LangGraph • Python • LLMs • Flask

- Architected a full agentic AI pipeline: document ingestion, knowledge-graph construction, LLM-powered retrieval, reasoning, and REST API serving.
- Implemented entity/relation extraction, Cypher generation, and grounding evaluation; exposed clean Flask API with tests and clear interfaces.

Real-Time Traffic Analysis System

Python • OpenCV • AWS

- Deployed vehicle detection model (~95% accuracy) to AWS production environment with ~30% latency reduction; integrated classical ML forecasting for congestion prediction.

AI-Powered Heart Disease Prediction

Python • scikit-learn • XGBoost

- Trained ensemble models achieving 90%+ accuracy; addressed class imbalance with PCA/SMOTE; benchmarked against neural network baseline.

EDUCATION

Master of Computer Science (Data Science & AI)

2024 – 2025

The University of Sydney • Sydney, Australia

B.Sc. Programming & Data Science (CGPA: 8.02) (online degree)

2020 – 2023

Indian Institute of Technology Madras • Chennai, India

B.Sc. Physical Science with Computer Science (CGPA: 8.77)

2020 – 2023

Hansraj College, University of Delhi • Delhi, India

PUBLICATIONS

- Goyal Y. et al. - Antiprotozoal Peptide Prediction using ML algorithms and feature selection. Published in Heliyon. Achieved significant reduction in false-positive rates.
- Goyal Y. et al. - iIL13Pred: Improved Prediction of IL-13 Inducing Peptides. Published in BMC Bioinformatics. 15% improvement in prediction accuracy over prior methods.

AWARDS & RECOGNITION

- Winner SAS Viya for Learners Challenge (ANZ), 2025: Top placement in regional analytics/ML competition.
- Winner Smart India Hackathon (SIH), 2023 · Winner HackLife 2023 (Devpost), 2023.
- Hackathon Mentor - Humanitarian Innovation Hackathon, USYD: mentored teams placing 1st & 2nd.